



Habib University
shaping futures

Data Science for Social Good

CS/SDP 312/314 L1

Fall Semester 2026

Course Information

Start Date	End Date	Class Location	Meeting Time
Aug 19 2026	Dec 08 2026	H-507	Mon Fri (11:40 AM-12:55 PM)

In God we trust; all others must bring data.

~.William Edwards Deming

Instructor Information

Instructor: Muhammad Usman Arif

Office Location: N-204

Email: usman.arif@sse.habib.edu.pk

Office Hours: Monday 9 am - 11 am

This course can be taken as a CS upper-level or an SDP upper-level elective and can be counted towards both CS and SDP Minor(s). Additionally, the course can be taken as a free elective, provided that the prerequisites of the course are met.

Course prerequisites include:

For CS: Math 310, CS 101

For SDP and other AHSS students: R language and SDP 202B (Quantitative Research Methods). The students must take all the prerequisites mentioned in their program to register for this course.

Course Description

This course introduces students to the core concepts and practices of data science, with a focus on making sense of real-world, messy data. Students will learn how to clean, organize, and analyze data to uncover patterns, trends, and insights. They'll explore both predictive models, which help anticipate future outcomes, and unsupervised techniques, which reveal natural groupings or structures within the data.

Designed to be hands-on and interdisciplinary, the course blends statistics, computing, and critical thinking. It emphasizes practical skill-building through the use of real datasets and widely used tools. In keeping with its social impact focus, the course integrates case studies aligned with the United Nations Sustainable Development Goals (SDGs), covering areas such as public health, education, climate, water security, and institutional governance. Through these cases, students will reflect on how data science can inform policy, improve social programs, and contribute to positive change.

By the end of the course, students will be able to work confidently with data, apply analytical techniques, and communicate findings that matter—whether to support decision-making, advance research, or address societal challenges.

Course Learning Outcomes (CLOs)

By the end of the course, students will be able to:

1. CLO: Develop skills in data manipulation, visualization, statistical analysis, and machine learning techniques
2. CLO: Implement and evaluate Data Science and Machine Learning algorithms to draw conclusions and predict outcomes based on real-world data.

3. CLO: Explore real-world applications of data science and recognize how machine learning techniques can support socially meaningful outcomes and responsible decision-
4. CLO: Analyze and interpret complex datasets using statistical methods and visualizations to extract and communicate meaningful insights.

Engagement & Participation Rules

Please read the campus safety policy and protocols if the classes are in-person.

You are expected to maintain a behavior befitting Yohsin and acknowledge the classroom as a place of learning, exploration, and experimentation. The University's standard policies on attendance, inclusivity, office hours, and academic integrity apply in this course.

Concerns regarding a score will be entertained by the instructor up to a week after the release of the score. Concerns raised later will not be entertained. Requests for grace marks for whatever reason will not be entertained and each such request will result in a penalty of 1% from the overall score.

Course Aims

Data science is a transformative field that combines critical thinking, analytical skills, and technology to extract meaning from data. Data is shaping every discipline, from business and healthcare to social sciences, humanities, and the arts.

This course is designed to empower students from diverse backgrounds to help them leverage data for meaningful insights. You will gain hands-on experience in data manipulation, visualization, and interpretation, making data work for your field of interest.

Mode of Instruction

Medium of Instruction: The class will meet twice a week for 75 minutes. While in-person sessions are preferred, we will adapt if circumstances require changes. You will be informed of any updates regarding the mode of instruction.

Programming & Tools: This course involves hands-on programming, with Python and R as the targeted languages. You will work with advanced libraries and methodologies commonly used in data science. These skills will be essential for homework assignments and your final project. Basic understanding of programming is a pre-requisite for the course. Tutorials for any advance programming concepts and tools will be shared as an additional resource and can be

explored on as per need basis. **Programming component will not be discussed during synchronous sessions.**

Final Project: You will collaborate preferably in an interdisciplinary team to design and implement a data science project of your choice based on real-world data. The project will require implementation and application of techniques and concepts learned throughout the course.

Course Resources & Communication: The Canvas LMS will serve as the main platform for sharing the syllabus, assignments, and course materials. Official communication will take place through Canvas, and it is your responsibility to stay updated. Please use the course site for discussions, clarifications, and course-related queries. All resources, including lecture slides, reference books, articles, additional support materials, and programming tutorials will be accessible through Canvas.

Assessments

Quiz (20%)	Theory Quizzes, in person, on paper
Homework Assignments (10%)	Take home, group-based, majorly programming-based
Class Participation (05%)	Class Participation through Q&A, polls, and discussions
Final Exam (30%)	Theory-based exam after week 16. In person, paper based, Duration will be 120 to 150 minutes.
Final Project (30%)	A comprehensive data science project to be taken in teams of upto 3 members. The project is divided into multiple milestones, each carrying separate marks. Proposal submission is expected in week 3. Final submission is due in week 16.
Project Manuscript and Poster (05%)	A manuscript and poster submission similar to a conference research submission related to the project attempted by the students.

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95]

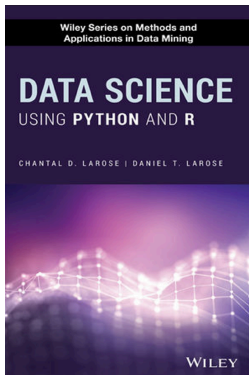
Letter Grade	GPA Points	Percentage
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60)

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Late Submission Policy

Late submissions after 24 hours will not be accepted. You will be penalized with 20% points for a late submission.

Required Texts and Materials

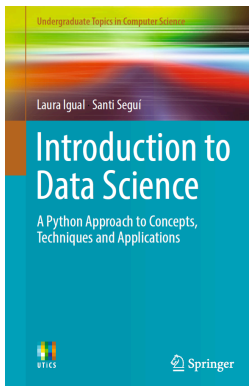


Data Science Using Python and R

ISBN: 9781119526834

Authors: Chantal D. Larose, Daniel T. Larose

Publisher: Wiley



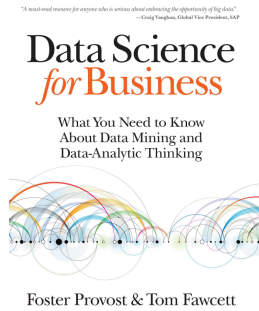
Introduction to Data Science

Subtitle: A Python Approach to Concepts Techniques and Applications

ISBN: 978-3-319-50016-4

Authors: Laura Igual · Santi Seguí

Publisher: Springer



Data Science for Business
ISBN: 978-1-449-36132-7
Authors: Foster Provost and Tom Fawcett
Publisher: O'REILLY

Attendance Policy

The University's standard Attendance Threshold is a minimum 85% of all sessions (i.e., students must be present in at least 85% of all classes/sessions of a given course) scheduled during the course of the semester. However, instructors may nominate a stricter attendance threshold (but cannot relax it more than the aforementioned 85%). Students must abide by these attendance thresholds set by the relevant faculty. If/Once a student's attendance falls within the aforementioned Attendance Threshold in a given course, they cannot receive credit for that course and will be removed from the course by the OASR. Students must inform their instructor within 48 hours after session to update the discrepant record on PSCS. So that the same may be corrected manually on the relevant attendance roster by the instructor. In cases where students are absent from a session on account of medical issues or family emergencies, they must inform/intimate the OASR within seven (7) days.

For more information, please refer to the Attendance Policy developed by OASR at Habib University.

Week-Wise Schedule (Tentative)

Fall 2025 Weekly Schedule

Week	Topic(s)	Activities/ Deliverables	Comments
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Week 1 August 19 - 21 Starting from Wed.	Introduction to the course. Syllabus walk-through Programming I (Programming resources and platform introduction)		Additional programming review material for Python/R released
Week 2 August 24 - 28 † 12th Rabi-ul Awal	Understanding Data and its Attributes		Basic Data Handling in Python/R material shared
Week 3 Aug 31 - Sep 4	Descriptive Statistics Review and Exploratory Data Analysis (EDA)	Quiz 1 (Programming Basics) HW 1 release	Programming reference material for data analytics and visualization shared https://www.kaggle.com/code/stephaniemozley/mental-health-dataset-data-quality-eda
Week 4 Sep 7 - 11	Inferential Statistics Review Programming II (EDA in Python and R)	Project: 01. Problem Framing and Dataset Curation (Team Formation) (4 marks)	
Week 5 Sep 14 - 18	Introduction to Machine Learning	Quiz 2 HW 1 Submission	Reference material for ML libraries in Python/R released
Week 6 Sep 21 - 25	Model Evaluation Metrics	HW 2 release Quiz 3	https://archive.ics.uci.edu/dataset/529/early+stage+diabetes+risk+prediction+dataset

Week 7 Sep 28 - Oct 2	KNN	Project: 02. EDA and Hypothesis Report (5 Marks)	https://www.kaggle.com/datasets/zeeshier/weather-forecast-dataset/data
Week 8 Oct 5 - 9	Holdout and Cross-validation Bias in Training Data Programming III (ML in Python and R)	HW 2 Submission	
Week 9 Oct 12 – 16	Logistic Regression		
Mid-Semester Recess Oct 17-20			
Week 10 Oct 21 - 23	Regression analysis	Project: 03. Baseline Model Submission and Initial Findings (5 marks)	https://www.kaggle.com/datasets/atharvasoundankar/global-water-consumption-dataset-2000-2024/data
Week 11 Oct 26 - 30	Neural Networks	Quiz 4	https://www.kaggle.com/datasets/uom190346a/water-quality-and-potability/data

Week 12 Nov 2 - 6	Ensemble models	Project: 04. Final Model, Fairness and Ethics Modeling (6 Marks)	https://archive.ics.uci.edu/dataset/320/student+performance
Week 13 Nov 9 - 13	Clustering Introduction	Quiz 5	https://www.kaggle.com/datasets/khushikyad001/corruption-perception-data
Week 14 Nov 16 - 20	Hierarchical Clustering	Project: 05. New Data Integration and Re-training Analysis (5 Marks)	
Week 15 Nov 23 - 27	Ethics and Fairness in Data Science (Bias in Data and Models, Fairness, Accountability)		https://www.kaggle.com/datasets/rohan0301/unsupervised-learning-on-country-data/data
Week 16 Nov 30 - Dec 8	Final Project Presentations	Project: 06. Presentations (5 Marks) Poster and Manuscript Submission (5 Marks)	
Week 17 Dec 10 - 16	Final Exam		

* The University reserves the right to correct typographical errors or to adjust the Academic Calendar at any time it deems necessary.

† Subject to the sighting of the moon.

‡ No Class(es).

Final Exam Policy

The final exam will be an in-person, on-paper exam during the final exam week.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, // bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
- c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never

involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

Some of the homework and final project will be given as a group activity. Unless mentioned, collaboration other than for group activities is not allowed.

Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

- PLO 1: Theoretical Computer Science: recall and apply foundational principles of computer science.
- PLO 2: : Application Development: build software systems of varying complexity in light of fundamental computer science principles and any other constraints.

- PLO 3: Analysis and Design: perform technical analysis and design using core computing and mathematical knowledge.
- PLO 4: Systems: apply the knowledge of computing systems.
- PLO 5: : Research and Exploration: develop expertise in and contribute to a given sub-field of computing by drawing upon a strong foundation in the fundamentals of computer science and mathematics to solve real-life problems.
- PLO 6: Problem Solving: identify and analyze problems and propose effective computing-based solutions.
- PLO 7: Practical Exposure: make effective use of current tools, technologies, and good industry practices.
- PLO 8: Responsible Citizenship: conduct their computing practice in a manner that is ethical and socially responsible and corresponds to their distinct sense of identity and service to the community.
- PLO 9: Self-Learning: continuously adapt their skills to the changes taking place around them.
- PLO 10: Design Thinking: apply design thinking principles to the design of a solution.
- PLO 11: Multi-disciplinarity: incorporate knowledge and input from multiple disciplines.
- PLO 12: Communication and Teamwork: communicate and function effectively as a member or a leader of a variety of teams.

Computer Science - Program Learning Outcomes (PLOs) mapped to Course Learning Outcomes (CLOs)				
	PLO 03	PLO 05	PLO 06	PLO 07
CLO 01	X	X		
CLO 02	X	X		X
CLO 03		X	X	
CLO 04		X	X	X

SDP - Program Learning Outcomes (PLOs) mapped to Course Learning Outcomes (CLOs)			
	PLO 01	PLO 02	PLO 04
CLO 01	X	X	X
CLO 02	X		
CLO 04	X	X	X

Mapping of Assessments to CLOs

Mapping

Assessment	CLO 01	CLO 02	CLO 03	CLO 04	Topic Details
HW 01	x	x			EDA and Visualization using Python/R
HW 02		x	x	x	Implementation of a basic ML model using Python/R Libraries and reporting results
Quiz 01	x				Data Basics
Quiz 02	x	x			EDA and Statistics
Quiz 03			x		Decision Trees
Quiz 04			x	x	Model Performance, Evaluation, Cross Validation, Holdout
Quiz 05			x	x	Prediction and Ensemble Models
Final Project	x	x	x	x	
Manuscript and Poster	x	x	x	x	
Final	x	x	x		

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues

- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.